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New age, geochemical and paleomagnetic data on a 2.21 Ga dyke swarm from south India: Constraints on Paleoproterozoic reconstruction

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ABSTRACT

Geochemical, paleomagnetic and Sm–Nd whole rock-mineral isochron age data are presented for a ~450 km-long N-S striking mafic dyke swarm from the Dharwar craton, southern India. Dykes in this swarm are basaltic tholeiite in composition and despite traversing diverse Archean rocks display limited compositional heterogeneity along their entire outcrop. Two whole rock-mineral Sm–Nd isochrons from one of these dykes yield ages of 2173 ± 43 Ma and 2190 ± 51 Ma (2σ) respectively. These ages are consistent with recently reported U–Pb ages of 2209.3 ± 2.8 Ma and 2220.5 ± 4.9 Ma for two other dykes within the craton that show a similar trend and chemical composition (French and Heaman, 2010), suggesting emplacement of this swarm close to 2210 ± 10 Ma. Paleomagnetic directions obtained from fifteen sites along the length of this swarm are all similar within errors despite dissimilar orientation between dykes towards the northern end of the swarm, with a grand mean of D_m = 239° and I_m = −64°, corresponding to a paleopole at lat. 32°S; long. 302°E (dp = 8, dm = 10). This Paleoproterozoic pole from the Dharwar has coeval poles from the Slave, Superior and Rae provinces and allows a Paleoproterozoic reconstruction of these cratons. Our reconstruction at ~2210 Ma shows that all these cratons could have been located within about 30° of each other, with Slave and Dharwar being remarkably closer (<10°). However, lack of reliable Neoarchean paleomagnetic data from the Dharwar inhibits tracing its ancestry to either of the Neoarchean Supercratons: Sclavia or Superia.

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1. Introduction

Earth's history records numerous periods of accentuated mafic magmatism. Those magmas genetically unrelated to seafloor spreading and subduction are the Large Igneous Provinces (LIPs) and include continental flood basalts. Proterozoic LIPs are generally deeply eroded to expose giant dyke swarms, several of them being of continental dimensions. Recognition of coeval giant dyke swarms on different continents could be extremely useful for continental reconstructions, for deciphering the frequency of mantle perturbation events and also for understanding the chemical evolution of the mantle through Proterozoic.

Like in most Archean cratons, the Dharwar craton was also invaded by a large number of mafic dyke swarms, which are predominantly of Paleoproterozoic age (for recent summaries: Halls et al., 2007; French and Heaman, 2010), apart from those of Cretaceous age (Kumar et al., 1988, 2001; Radhakrishna et al., 1994). Of these, the recently discovered E-W to ENE-WSW trending giant radiating dyke swarm emplaced at 2367 Ma with an areal extent of nearly the entire eastern Dharwar craton appears to be the most dominant (Kumar et al., 2012). Other dyke swarms that have been dated recently using U–Pb baddeleyite geochronology from the eastern Dharwar craton include a N-S oriented swarm at 2220.5 ± 4.9 Ma, a NW-SE striking swarm at 2209.3 ± 3.8 Ma and a radial WNW-ESE to NW-SE swarm at 2180.8 ± 0.9 to 2176.5 ± 3.7 Ma (French and Heaman, 2010). The N-S striking dyke swarm appears to extend for over 450 km and straddle nearly the entire length of the Dharwar craton (Fig. 1). In order to establish the spatial extent of this dyke swarm, lateral variation in composition and study their paleomagnetic signature, a concerted geochemical, paleomagnetic and geochronological study was undertaken. Based on these results, we also attempt here a Paleoproterozoic continental reconstruction involving the Dharwar craton.

2. Regional geology and sampling

The Archean Dharwar craton as summarized by Swami Nath and Ramakrishnan (1981), Drury et al. (1984) and Naqvi and Rogers (1987) consists of four blocks. These are the Southern granulite block, the Eastern Ghats granulite block and the low-grade Northern block that is further divisible into western and eastern blocks based on contrasting metamorphic *P–T* conditions. The latter two

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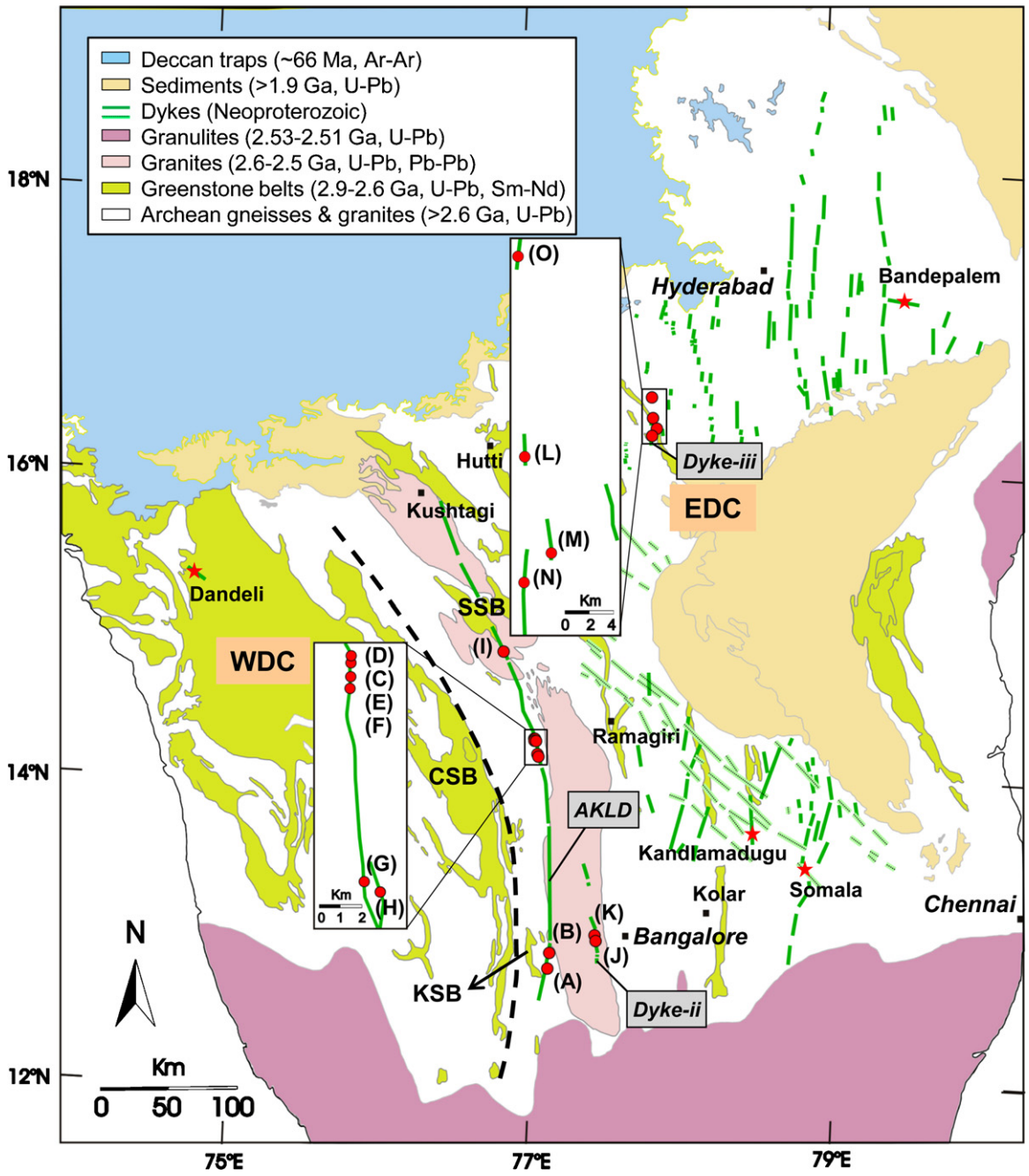


Fig. 1. Simplified geology map (modified after Naqvi and Rogers, 1987, dykes traced from Google earth and French and Heaman, 2010) of a part of the Dharwar craton showing AKLD (Andhra Karnataka long dyke) dykes ii and iii, other ~2210 Ma dykes (after French and Heaman, 2010). Red dots indicate sampled site locations of this study, A–I in parentheses are sites on AKLD, J and K on dyke ii and L–O on dyke iii. Red stars in the map indicate site locations of samples used for U–Pb age determinations on the Kandlamadugu (2220.5 ± 4.9 Ma), Somala (2209.3 ± 2.8 Ma) and 2181–2176 Ma Bandepalam-Dandeli dykes by French and Heaman (2010). EDC and WDC are Eastern Dharwar Craton and Western Dharwar Craton respectively, SSB: Sandur schist belt, CSB: Chitradurga schist belt, KSB: Kunigal schist belt. Black dashed-line demarcates the boundary between the Eastern and Western Dharwar craton.

blocks are separated by a north-south fault zone proximal to the eastern margin of the Chitradurga belt (Fig. 1).

Of the three dykes sampled in this study one of the dykes is unusually long and runs nearly parallel to the eastern margin of the Chitradurga schist belt to its east for well over 350 km, along the western edge of the eastern Dharwar craton. In its southern most part, east of the Kunigal schist belt, the dyke intrudes mid-Archaean amphibolite grade gneisses, while along its middle and northern exposure it cuts late-Archaean K-rich granites, low-grade gneisses and schists such as the Sandur schist belt (Fig. 1). The strike of the dyke is generally NNE in the southern and N-S in the middle parts, but swings ~25° NNW in its northern half, thus broadly following the regional structural grain of the basement rocks. Thickness of the dyke is irregular along its length (shows no systematic variation in thickness along strike), and varies between 40 and 230 m. The dyke preserves a vertical dip along the entire strike length sampled. It is medium to coarse grained and has sharp contacts with the country rock. Two of the sampling sites located along the southern most part of the dyke are in the amphibolite region of the craton whereas the six sites in the middle portion and one site in the northern part of the dyke are located in the greenschist facies regions (Fig. 1). The increasingly deeper exposure level of the crust from north to

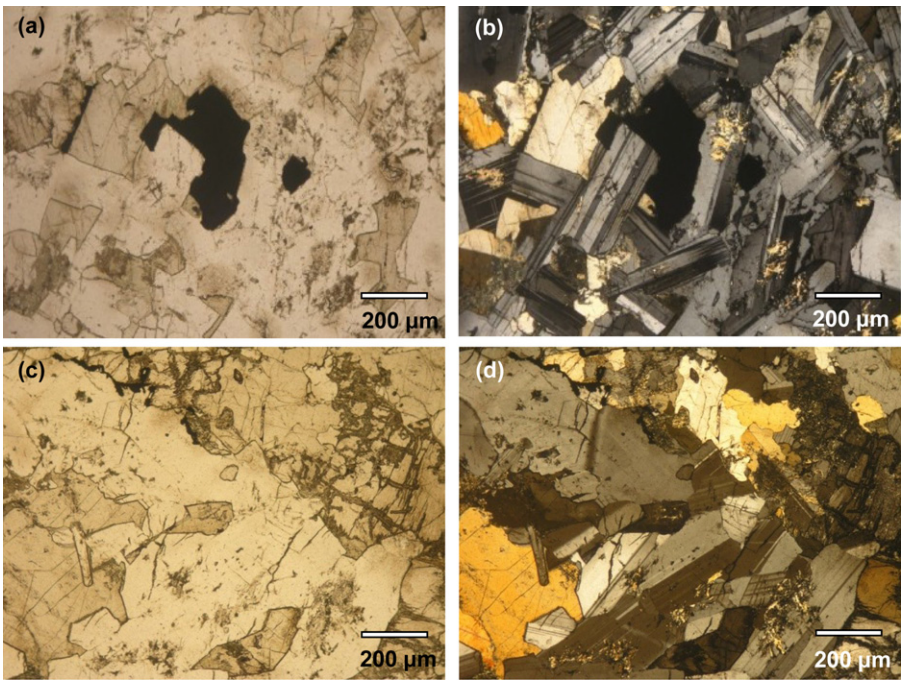


Fig. 2. Photomicrographs of AKLD samples. Samples are characterised by minimum alteration. (a) and (b) are from site A (country rock metamorphosed to amphibolite grade), the southernmost site on the AKLD and (c) and (d) from site F (intruded into greenschist facies rocks) from one of the northern sites on it.

south is believed to be due to the 2–5° northward tilt of the Dharwar craton (Harris and Jayaram, 1982). Sampling could not be done evenly all along the dyke length due to paucity of fresh in situ outcrops. The studied dyke, though not the largest of the giant dykes known elsewhere, is clearly the longest continuously exposed dyke (from east of the Kunigal schist belt and extending up to Kushtagi in the north, Fig. 1) in the Dharwar craton, running across the political boundaries of two large south Indian states, Andhra Pradesh and Karnataka. We herein name this geological entity as Andhra-Karnataka Long Dyke (AKLD) after the two states. Besides the AKLD (nine sites, A–I) we have also sampled two other N-S striking dykes (dyke ii and dyke iii) both east of it, dyke ii (two sites, J and K) is located about 50 km from sites A and B and dyke iii (four sites, L, M, N, O) about 200 km northeast of site I (locations of dyke ii and dyke iii are given in Fig. 1).

3. Experimental procedures

At least one sample from each of the nine sites on the AKLD and on the other two dykes was analyzed for major and trace element compositions (including Rare Earth Elements-REE) to help understand the variation in geochemistry along the strike of this dyke swarm. All the geochemical analysis were done at the NGRI using a wave length dispersive XRF for major elements on pressed-pellets, trace and REE by a quadrupole ICP-MS. Details of the analytical procedures are given in Krishna et al. (2007) and Roy et al. (2007) respectively.

Oriented block samples were collected using Brunton magnetic compass. At least three cores of 2.56 cm diameter were later drilled from each of the block samples and these were cut into cylinders of 2.2 cm length, resulting in a minimum of five cylindrical specimens from each block sample and 290 specimens in total. Almost all the samples for this study were collected from fresh exposures in streams, rivulet beds, road cuttings or quarries. Sample susceptibility and Curie temperature runs were conducted on Bartington Instruments model MS-2. Isothermal remnant magnetization (IRM) studies were performed on an ASC Scientific Model IM-10–30 impulse magnetizer. Standard demagnetization procedures including incremental alternating field (AF) and heat treatment (TH) were performed using Molspin Alternating Field demagnetizer and MMTD 80 oven (Magnetic Measurements Limited) respectively. After each step of incremental demagnetization, the natural remanent magnetization (NRM) was measured using either Molspin or JR-6 spinner magnetometer (Agico).

For Sm–Nd analysis, each of the two 250 g samples (HD-14 and HD-19, both from AKLD) of the dyke were crushed separately to less than 5 mm fragments, homogenized and split into two portions (1:4). The smaller split was finely ground (to <400 mesh) for whole rock analysis. The larger split was crushed and sieved to 100–150 μm size fraction, which was density separated in bromoform and methylene iodide into mainly plagioclase (ρ 2.65–2.75 g/cm^3) and pyroxene (ρ 3.3 g/cm^3) separates. These two density fractions were handpicked under a binocular microscope to obtain 99% purity, while magnetite was separated using a hand magnet from the pyroxene-rich fraction. All the fractions were acid leached in 0.2 M HCl for 15 min before dissolution.

Whole rocks and mineral separates were dissolved in distilled $HF + HNO_3$ and split into two portions for isotopic composition and concentration measurements, respectively. Sm and Nd were separated using two ion exchange columns (Kumar et al., 1998). Isotopic measurements were made on VG-354 five collectors Thermal Ionization Mass Spectrometer with all ions as metals, at the NGRI (Kumar et al., 1999). Total procedure blanks for Sm and Nd are < 30 pg. La Jolla Nd standard analyzed during this study gave an $^{143}Nd/^{144}Nd$ ratio of 0.511864 ± 14 (2σ, weighted mean of six runs, using $^{146}Nd/^{144}Nd = 0.7219$ for mass fractionation correction). ^{147}Sm and ^{146}Nd spikes were calibrated against La Jolla mixed Sm–Nd standard. Quoted uncertainties for all the data presented here are 2σ external precision level based on replicate measurements of standards during the course of this work. Isochron fitting and age computations were done using ISOPLOT software (Ludwig, 2003). Regression errors are quoted at 2σ level. The ^{147}Sm decay constant used is 6.54×10^{-12} year^{-1} (Steiger and Jager, 1977).

Table 1
Dharwar 2.21 Ga N-S dyke swarm geochemical data.

Sample (Site)	HD12[c] (A)	HD14[c] (A)	HD86/19[c] (B)	GD6[d] (C)	GD6a[c] (C)	GD8[d] (D)	GD10[d] (D)	GD12[c] (E)
wt%								
SiO_2	48.49	49.03	49.10	48.57	48.28	49.06	48.69	48.42
Al_2O_3	14.79	14.19	13.54	13.10	14.11	14.16	14.06	14.16
Fe_2O_3	13.13	14.07	14.02	14.09	14.54	14.02	15.04	14.22
MnO	0.13	0.14	0.16	0.15	0.11	0.12	0.12	0.12
MgO	6.77	6.24	6.36	6.91	6.27	5.35	5.24	6.58
CaO	9.87	10.00	10.11	10.02	10.17	10.03	9.93	10.18
Na_2O	2.67	2.48	2.44	2.60	2.69	2.91	2.73	2.60
K_2O	1.07	1.14	1.06	1.11	1.17	1.28	1.28	1.09
TiO_2	1.30	1.37	1.47	1.51	1.53	1.52	1.73	1.26
P_2O_5	0.08	0.10	0.08	0.10	0.11	0.13	0.12	0.08
Total	98.30	98.76	98.34	98.16	98.98	98.58	98.94	98.71
ppm								
Sc	40.48	44.05	47.79	40.63	32.64	34.61	34.65	36.53
V	299.56	329.11	318.94	343.61	324.55	326.88	348.66	270.44
Cr	168.75	215.49	194.78	211.05	161.56	56.81	56.98	156.91
Co	101.42	55.20	60.27	81.57	47.28	69.12	45.83	48.26
Ni	139.76	148.56	160.23	139.45	115.09	75.41	66.78	126.03
Cu	100.18	100.16	118.83	90.58	107.96	95.40	94.45	92.19
Zn	89.34	99.97	100.59	80.94	96.33	100.06	104.22	83.71
Ga	17.03	17.11	16.95	16.95	19.31	20.02	20.52	17.68
Rb	12.26	16.13	12.35	16.40	19.21	20.25	21.21	13.18
Ba	75.73	90.43	75.50	86.82	109.01	128.96	131.81	85.96
Th	1.10	1.35	1.04	1.54	1.82	2.10	2.19	1.25
U	0.30	0.42	0.30	0.43	0.50	0.67	0.61	0.37
Nb	2.47	3.04	2.73	2.90	3.59	3.95	4.37	2.56
Ta	0.97	0.44	0.45	0.49	0.50	0.58	0.45	0.41
La	5.33	6.30	5.30	6.80	7.98	9.72	9.03	5.90
Ce	11.91	14.41	11.95	15.29	17.58	21.45	20.33	13.28
Pr	1.54	1.89	1.55	1.97	2.19	2.68	2.61	1.70
Sr	162.24	157.13	149.78	149.01	173.72	174.19	173.48	166.17
Nd	6.96	8.45	6.96	8.73	9.74	11.93	11.52	7.39
Zr	53.13	63.10	45.41	59.92	72.22	86.98	83.99	51.08
Hf	1.34	1.61	1.28	1.65	1.80	2.16	2.14	1.37
Sm	1.91	2.38	1.93	2.44	2.64	3.21	3.10	2.08
Eu	0.68	0.79	0.70	0.80	0.91	1.04	1.05	0.72
Gd	2.20	2.76	2.30	2.86	3.01	3.56	3.65	2.34
Tb	0.40	0.49	0.42	0.51	0.52	0.64	0.65	0.42
Dy	2.66	3.20	2.79	3.39	3.33	4.16	4.23	2.71
Y	17.15	20.94	18.17	21.38	21.09	26.29	26.89	17.67
Ho	0.56	0.69	0.58	0.70	0.70	0.86	0.87	0.58
Er	1.62	1.95	1.72	2.06	1.98	2.49	2.51	1.69
Tm	0.26	0.33	0.28	0.32	0.31	0.40	0.40	0.26
Yb	1.60	1.94	1.65	1.96	1.85	2.41	2.43	1.62
Lu	0.24	0.29	0.25	0.30	0.28	0.36	0.37	0.25
$\sum$(REE)	37.86	45.84	38.37	48.14	53.01	64.90	62.74	40.95
Nb/Th	2.25	2.25	2.63	1.88	1.97	2.17	2.00	2.05
La_N/Sm_N[a]	1.74	1.65	1.72	1.74	1.89	1.89	1.82	1.77
Sm_N/Yb_N[a]	1.30	1.34	1.27	1.36	1.55	1.45	1.39	1.40
La_N/Lu_N[a]	2.34	2.24	2.18	2.38	2.91	2.77	2.56	2.42
Nb_N/La_N[b]	0.46	0.48	0.51	0.42	0.44	0.40	0.48	0.43

Sample (Site)	GD13[c] (E)	GD15[c] (F)	GD20[c] (G)	GD24[c] (H)	GD25[c] (H)	GD26[c] (I)	GD29[c] (I)	GD29a[c] (I)
wt%								
SiO_2	49.02	47.59	48.54	48.74	48.47	48.19	48.30	47.44
Al_2O_3	13.21	13.06	13.27	15.03	14.19	13.43	13.02	13.52
Fe_2O_3	13.15	14.38	14.32	13.30	13.51	13.74	13.65	14.13
MnO	0.14	0.20	0.12	0.12	0.15	0.18	0.18	0.20
MgO	7.43	8.24	6.71	7.14	7.03	7.40	8.22	8.14
CaO	10.59	10.22	10.01	10.07	10.41	10.37	10.40	10.27
Na_2O	2.39	2.28	2.65	2.50	2.12	2.43	2.24	2.15
K_2O	1.03	0.96	1.07	1.05	1.19	1.08	1.08	1.11
TiO_2	1.38	1.43	1.29	1.23	1.30	1.55	1.45	1.71
P_2O_5	0.06	0.07	0.07	0.07	0.09	0.09	0.08	0.09
Total	98.40	98.43	98.05	99.25	98.46	98.46	98.62	98.76
ppm								
Sc	40.80	55.07	39.69	36.31	43.90	48.29	46.41	50.42
V	336.63	321.77	300.18	252.34	276.77	318.58	341.08	367.45
Cr	200.36	133.01	164.40	186.94	195.28	231.55	242.11	248.61
Co	54.76	68.93	50.98	48.77	78.74	81.73	78.01	65.97
Ni	159.48	174.17	132.83	139.38	148.43	169.54	162.75	165.24
Cu	122.05	156.35	99.01	95.03	103.62	121.64	107.80	132.15

Table 1 (*Continued*)

Sample (Site)	GD13[c] (E)	GD15[c] (F)	GD20[c] (G)	GD24[c] (H)	GD25[c] (H)	GD26[c] (I)	GD29[c] (I)	GD29a[c] (I)
Zn	89.89	101.17	87.62	77.07	97.05	106.74	102.59	116.25
Ga	15.60	15.56	17.27	17.06	16.92	16.25	16.20	17.03
Rb	11.45	9.21	12.12	12.75	18.19	12.43	15.68	18.30
Ba	63.70	57.33	83.19	75.32	89.97	82.64	77.15	82.97
Th	0.93	1.14	1.23	1.17	1.31	1.43	1.31	1.52
U	0.27	0.32	0.36	0.33	0.39	0.45	0.37	0.45
Nb	2.10	2.37	2.50	2.15	2.65	2.87	2.74	3.35
Ta	0.28	0.41	0.41	0.32	0.56	0.53	0.51	0.36
La	4.31	5.11	5.41	4.86	6.30	6.01	5.85	6.64
Ce	9.67	11.52	12.10	10.92	14.07	13.53	13.21	15.03
Pr	1.26	1.51	1.55	1.41	1.81	1.77	1.69	1.98
Sr	140.30	119.85	166.32	171.26	159.23	136.38	138.96	128.30
Nd	5.67	6.78	7.06	6.28	7.95	7.87	7.59	8.77
Zr	39.94	47.87	50.84	38.23	54.06	58.39	40.85	57.47
Hf	1.12	1.23	1.40	0.96	1.46	1.57	1.12	1.42
Sm	1.62	1.92	1.96	1.76	2.15	2.21	2.20	2.42
Eu	0.58	0.66	0.73	0.63	0.76	0.74	0.74	0.78
Gd	1.88	2.28	2.27	1.99	2.47	2.59	2.46	2.82
Tb	0.35	0.42	0.40	0.36	0.46	0.48	0.45	0.50
Dy	2.24	2.83	2.67	2.42	2.97	3.09	3.00	3.39
Y	14.58	18.07	17.31	15.20	19.41	20.16	19.40	21.89
Ho	0.47	0.59	0.56	0.50	0.63	0.66	0.63	0.72
Er	1.39	1.73	1.63	1.44	1.84	1.90	1.84	2.11
Tm	0.22	0.28	0.27	0.23	0.29	0.31	0.29	0.34
Yb	1.37	1.67	1.60	1.44	1.76	1.90	1.75	2.02
Lu	0.21	0.27	0.24	0.21	0.27	0.28	0.26	0.30
$\sum$(REE)	31.22	37.55	38.44	34.45	43.74	43.35	41.96	47.81
Nb/Th	2.26	2.08	2.03	1.84	2.02	2.01	2.09	2.20
La_N/Sm_N[a]	1.66	1.67	1.73	1.72	1.83	1.70	1.66	1.71
Sm_N/Yb_N[a]	1.28	1.25	1.33	1.33	1.33	1.27	1.37	1.30
La_N/Lu_N[a]	2.17	2.00	2.33	2.38	2.40	2.21	2.32	2.28
Nb_N/La_N[b]	0.48	0.46	0.46	0.44	0.41	0.47	0.46	0.50

Sample (Site)	SD1[c] (J)	SD6[c] (K)	MB8[c] (O)	Means std. MRG-1	$\pm 2\sigma$
wt%					
SiO_2	48.74	48.18	49.30	39.41	0.48
Al_2O_3	13.92	13.65	14.04	8.35	0.10
Fe_2O_3	14.03	13.86	14.53	17.40	0.89
MnO	0.13	0.16	0.13	0.16	0.00
MgO	6.35	7.55	8.19	14.21	0.73
CaO	10.1	10.25	8.01	14.29	0.43
Na_2O	2.61	2.34	2.07	0.78	0.06
K_2O	1.14	1.08	1.74	0.30	0.11
TiO_2	1.45	1.42	1.34	3.90	0.26
P_2O_5	0.10	0.08	0.11	0.07	0.01
Total	98.57	98.57	99.46	–	–

Sample (Site)	SD1[c] (J)	SD6[c] (K)	MB8[c] (O)	BCR-1	$\pm 2\sigma$
ppm					
Sc	39.13	45.73	28.81	35.34	3.14
V	322.04	311.17	198.37	431.88	26.05
Cr	158.08	200.27	563.92	16.46	1.47
Co	62.63	67.59	54.80	38.03	1.76
Ni	125.64	156.05	169.72	13.30	0.72
Cu	102.42	116.51	97.56	19.36	0.42
Zn	93.89	98.36	45.12	132.89	4.26
Ga	17.91	16.61	13.95	22.66	0.62
Rb	15.83	14.1	24.84	47.97	0.98
Ba	94.21	78.37	405.10	680.01	5.22
Th	1.48	1.30	2.86	5.70	0.32
U	0.43	0.38	–	1.75	0.04
Nb	3.08	2.66	4.38	14.19	0.14
Ta	0.51	0.44	0.28	0.82	0.04
La	6.74	5.74	14.25	24.76	0.39
Ce	15.1	12.91	29.82	53.19	0.76
Pr	1.93	1.67	3.81	6.71	0.06
Sr	160.67	145.76	368.81	339.40	10.44
Nd	8.59	7.47	14.62	28.32	0.48
Zr	61.75	49.67	74.11	192.25	1.54
Hf	1.61	1.31	1.91	4.87	0.11
Sm	2.37	2.09	3.09	6.57	0.21
Eu	0.81	0.72	1.00	1.93	0.04
Gd	2.73	2.41	3.28	6.58	0.19
Tb	0.49	0.44	0.48	1.04	0.02

Table 1 (*Continued*)

Sample (Site)	SD1[c] (J)	SD6[c] (K)	MB8[c] (O)	BCR-1	$\pm 2\sigma$
Dy	3.19	2.91	3.18	6.25	0.10
Y	20.46	18.78	18.87	38.76	0.68
Ho	0.67	0.61	0.63	1.25	0.02
Er	1.93	1.78	1.65	3.62	0.07
Tm	0.31	0.29	0.24	0.55	0.01
Yb	1.87	1.73	1.58	3.33	0.06
Lu	0.28	0.26	0.24	0.50	0.01
$\sum$(REE)	47.01	41.03	77.85	–	–
Nb/Th	2.08	2.05	1.53	–	–
La_N/Sm_N[a]	1.78	1.72	2.88	–	–
Sm_N/Yb_N[a]	1.38	1.31	2.14	–	–
La_N/Lu_N[a]	2.50	2.29	6.16	–	–
Nb_N/La_N[b]	0.45	0.46	0.30	–	–

[a] Values normalizated to C1 chondrite.
[b] Values normalizated to primitive mantle (McDonough and Sun, 1995). Site locations are given as red dots in Fig. 1.
[c] Samples from margin, with in 50 cm from the dyke contact with the country rock.
[d] Samples beyond 50 cm from the dyke contact.

4. Results

4.1. Petrology and geochemistry

Petrographic studies have been carried out on at least one sample from each site. Samples are medium to coarse grained towards the centre of the dyke and relatively finer towards the margin. Plagioclase, augite, and occasionally hypersthene are the major mineral constituents with minor amounts of magnetite. Modal abundance of magnetite is 4–5%. Poikilitic texture is very common, though in a few sections porphyritic texture was also observed. All samples are generally fresh, barring minor alteration of plagioclase in a few samples. Degree of alteration or feldspar clouding does not appear to vary along the dyke strike (Fig. 2), though the degree of metamorphism in the country rock increases from north to south.

For geochemical characterization, compositional variation along strike and comparison with other well dated Paleoproterozoic dyke swarms in the Dharwar craton, major, trace and rare earth element compositions were obtained on nineteen dyke samples spread along its strike (Table 1). Majority of the analyzed samples represent dyke margins. All dyke samples regardless of age plot along the tholeiitic trend on the ternary AFM diagram (Fig. 3a; Irvine and Baragar, 1971), with AKLD, other N-S dykes of this study and Kandlamadugu forming a tight cluster, the Bangalore and Somala dykes have higher MgO concentrations, while the Bandepalem dykes have lowest MgO. Due to their moderate SiO_2 (47.4–49.3 wt%) and low total alkali compositions ($Na_2O + K_2O$ = 3.2–4.2 wt%), all samples plot in the basalt field (Fig. 3b; Le Maitre, 2002) except the older Bangalore dyke swarm samples, which plot in the basaltic andesite field due to higher SiO_2 compositions. This classification as basaltic tholeiite for the N-S dyke swarm samples is in good agreement with the mineral composition of these dykes summarized above. In the Harker diagrams (Fig. 4), all N-S dyke samples exhibit limited variation in major element concentrations. However, SiO_2 of Bangalore dykes, Al_2O_3, MgO, K_2O and P_2O_5 concentrations for the Bandepalem dyke are all inconsistent with the N-S dyke swarm compositions. To infer paleotectonic setting MgO–FeO–Al_2O_3 discrimination diagram of (Pearce et al., 1977) was used (Fig. 5). All N-S dyke samples plot on the boundary between continental and ocean island basalt fields. Bangalore, Somala, Bandepalem and Dandeli plot away from the N-S dyke cluster.

Normalized to average primitive mantle composition (Fig. 6a), all N-S dyke samples show similar patterns characterized by enrichments of large ion lithophile elements (LILEs) relative to high field strength elements (HFSEs) and by a significant negative-Nb anomaly. Sample from the northern most site 'O' (dyke iii) is relatively more enriched in LIL and HFS elements compared to other dykes of this study, but is similar to the Kandlamadugu dyke. Bandepalem and Dandeli dykes display relatively flat and distinctly different patterns compared to N-S dykes including AKLD, Kandlamadugu and Somala. On the chondrite-normalized REE plot (Fig. 6b), all samples from the N-S dyke swarm display enriched light rare earth elements (LREEs) with AKLD and dyke ii having similar $(La/Sm)_N$ of 1.65–1.89, $(Sm/Yb)_N$ of 1.25–1.55 and $(La/Lu)_N$

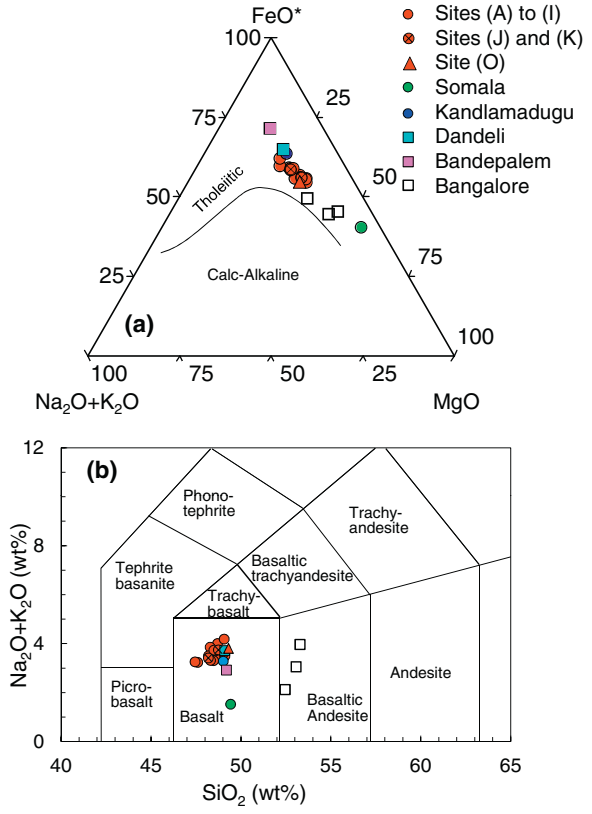


Fig. 3. (a) AFM diagram with discriminant boundary between tholeitic and calc-alkaline is from Irvine and Baragar (1971) and (b) total alkalis vs. silica diagram, boundary lines and fields as proposed by IUGS (Le Maitre, 2002) showing the compositions of dykes from this study and other U–Pb dated mafic dykes from the Dharwar craton. Kandlamadugu dyke (N-S striking, 2220.5 ± 4.9 Ma), Somala dyke (NW-SE striking, 2209.3 ± 3.8 Ma), Dandeli, Bandepalem (radial WNW-ESE to NW-SE swarm at 2180.8 ± 0.9 Ma to 2176.5 ± 3.7 Ma) and NE-SW 2367 Ma Bangalore dyke swarm data are from French and Heaman (2010).

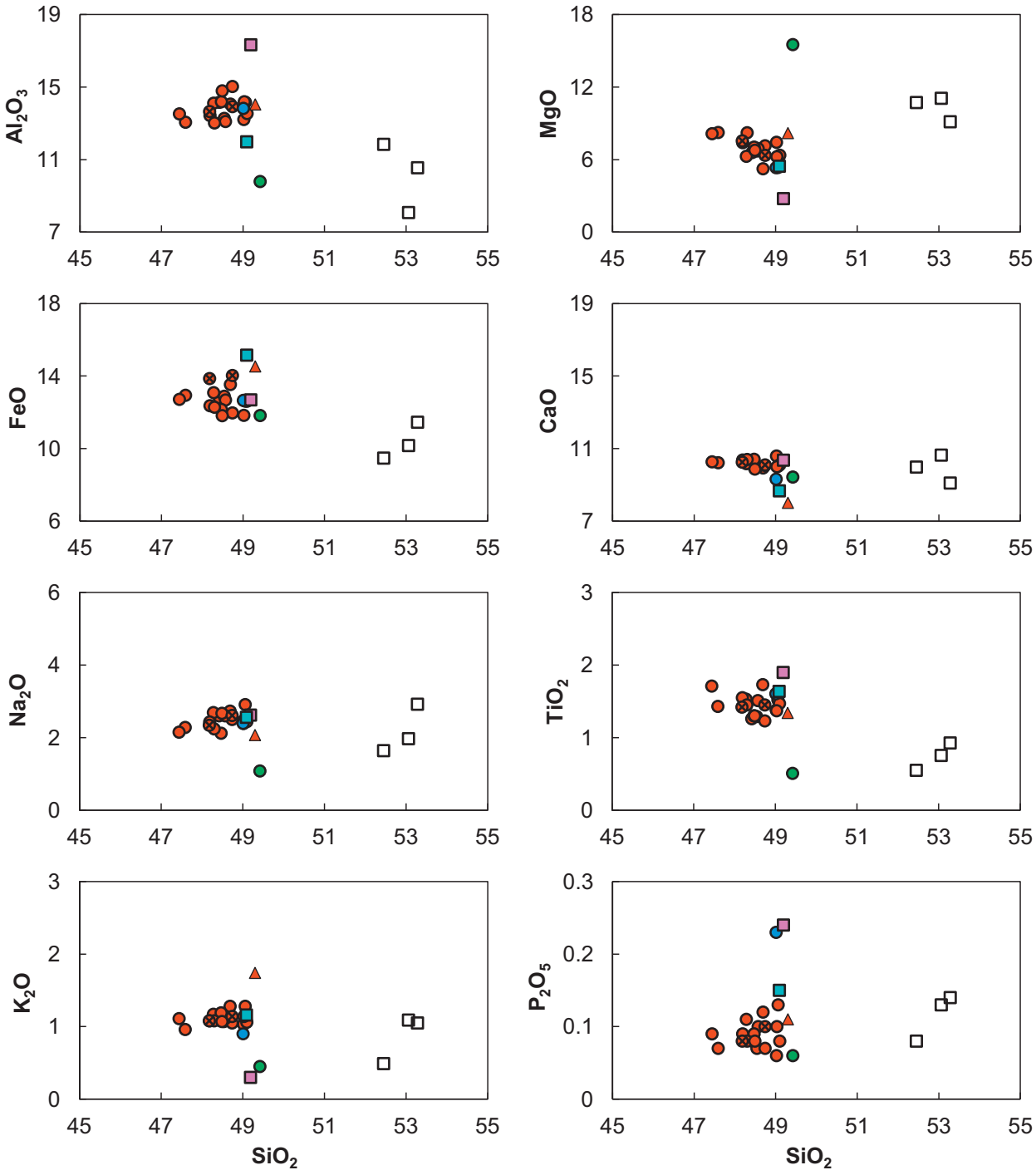


Fig. 4. Harker variation diagram for samples along AKLD, dykes ii and iii together with other U–Pb dated dykes (symbols as in Fig. 3a).

ratios of 2.00–2.91, with the total REE abundances not varying appreciably (31–65 ppm). Dyke iii (site O) and Kandlamadugu however have marginally higher total REEs of 78 and 73 respectively. The younger Bandepalem and Dandeli dykes having a relatively less fractionated REE pattern with their $(La/Lu)_N$ values being 1.59 and 1.27 respectively. Nb/Th ratio of mafic rocks was shown to be a sensitive indicator of crustal contamination (Jochum et al., 1991). Even minor amounts of crustal contamination, due to its elevated Th contents, will result in lowering Nb/Th values appreciably compared to estimates for primitive mantle (8.0; McDonough and Sun, 1995). All the samples investigated here have significantly lower Nb/Th ratios of 2.63–1.53, with site 'O' having the lowest value. Nb vs. Nb/Th and Th vs. Nb/Th plots (Fig. 7a and b) illustrate the compositions of these dykes, with respect to the primitive mantle. Similarly, Pearce (2008) demonstrated the utility of ratio–ratio plots such as Th/Yb vs. Nb/Yb for inferring the degree of crustal contamination and source composition. These trace element ratios for analyzed samples plotted in Fig. 8 tend to be displaced from the MORB-OIB array (Pearce, 2008) and towards average continental crustal compositions. Together Figs. 7 and 8 illustrate that these samples have suffered appreciable crustal contamination. However, despite traveling long distances (anisotropy magnetic susceptibility studies on AKLD, dyke ii and dyke iii samples suggest predominant horizontal magma flow all along their strike, A. Kumar and E. Nagaraju,

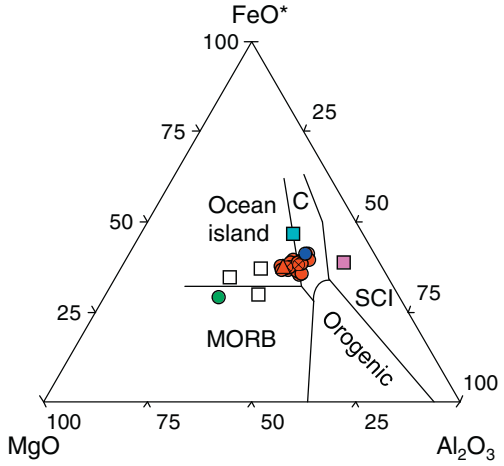


Fig. 5. MgO–FeO*–Al_2O_3 (after Pearce et al., 1977; * indicates total Fe) tectonic discrimination diagrams for AKLD, dykes ii and iii samples and other U–Pb dated dykes (symbols as in Fig. 3a).

manuscript in preparation, 2012) and traversing different continental crustal domains, all samples except site 'O' display fairly uniform contamination, with no systematic variation in elemental concentrations along strike in AKLD. This feature suggests that the observed contamination along AKLD may not have occurred during emplacement and residence in a crustal magma chamber, but that the magma feeding this and other dykes may have originated from a contaminated, enriched mantle source, but dyke iii at the northern extent of this swarm (having travelled longer distances) may have been affected by both.

To minimize the effects of fractionation, ratio–ratio plots of highly incompatible elements have been used to characterize mantle sources (Saunders et al., 1988). Following this approach samples of AKLD, dyke ii and dyke iii are plotted in (Fig. 9) the Th/Ta vs. Th/Tb bivariate diagram together with other U–Pb dated Paleoproterozoic Dharwar dykes. Distinct trends are clearly discernible for dykes of different ages indicating variegated differences in their source characteristics. Together, major, trace and REE characteristics of the U–Pb dated dykes suggest that Kandlamadugu dyke is chemically coherent with the AKLD and the other N-S dykes ii and iii studied here.

4.2. Sm–Nd geochronology

Sm–Nd analytical results are given in Table 2 and plotted in Fig. 10a and b. With limited spread in $^{147}Sm/^{144}Nd$ ratios for the pyroxene, plagioclase feldspar, magnetite and whole-rock for both the samples HD-14 and HD-19 when regressed separately yield ages of 2173 ± 43 Ma (MSWD-0.44) and 2190 ± 51 Ma (MSWD-0.11) respectively, and are similar within errors. Initial $^{143}Nd/^{144}Nd$

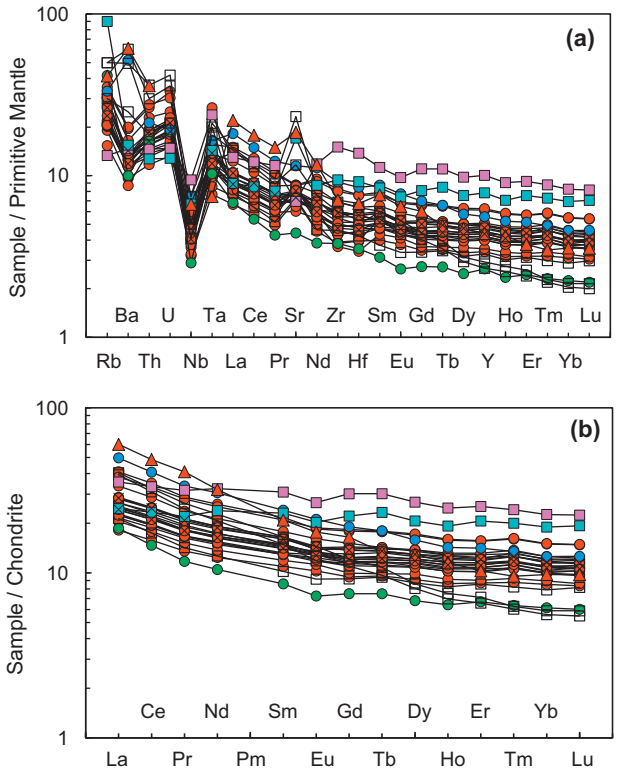


Fig. 6. (a) Primitive mantle-normalized incompatible trace elements spider diagram and (b) chondrite-normalized REE patterns for the dyke samples of this study and the published set of U–Pb dated dykes (symbols are the same as in Fig. 3a). All the normalizing values are from McDonough and Sun (1995).

Table 2
Sm–Nd data for samples HD 14 and HD 19.

Sample	Sm (ppm)	Nd (ppm)	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$
HD 14				
Fel-1	0.7715	3.343	0.1392	0.511730
Cpx	1.881	5.510	0.2059	0.512681
Fel-3	0.773	3.054	0.1526	0.511910
WR-1	1.514	5.128	0.1780	0.512278
Mag	0.7368	2.958	0.1502	0.511881
HD 19				
WR-1	1.968	6.899	0.1721	0.512132
Cpx-1	2.368	7.1893	0.1986	0.512512
Fel-1	0.872	3.622	0.1452	0.511740
Mag	1.176	4.265	0.1663	0.512050
Fel-6	0.663	2.777	0.1438	0.511724

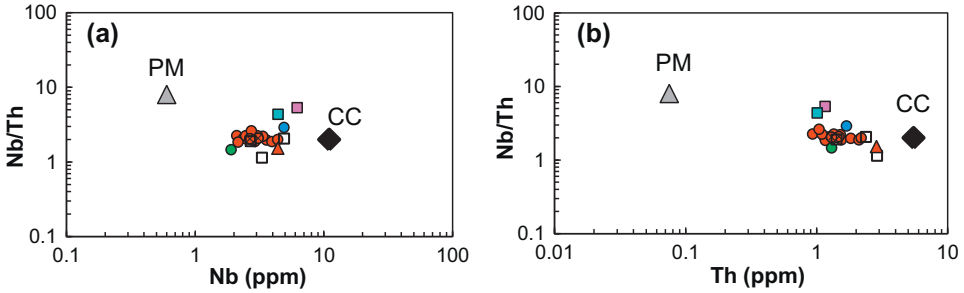


Fig. 7. (a) Nb/Th vs. Nb and (b) Nb/Th vs. Th plots for AKLD, dyke ii and dyke iii samples together with other U–Pb dated dykes (symbols are the same as in Fig. 3a). PM: primitive mantle (Hofmann, 1988); CC: crustal contamination (Taylor and McLennan, 1985).

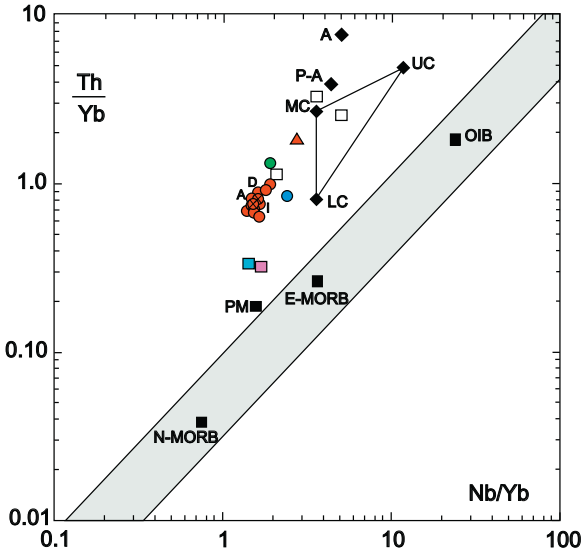


Fig. 8. Immobile incompatible element (Th–Nb–Yb) ratio diagram for AKLD, dyke ii and dyke iii with other U–Pb dated dykes (symbols as in Fig. 3a), after Pearce (2008). Deviation of dyke compositions above the MORB-OIB array suggests crustal contamination. Compositions for average felsic Archean (A) and post-Archean (P-A) crust, and for average lower (LC), middle (MC) and upper (UC) crust are taken from Rudnick and Fountain (1995). Estimates of N-MORB, E-MORB, ocean island basalt (OIB) and Primordial Mantle (PM) are taken from Sun and McDonough (1989). Shaded field represents MORB-OIB array, after Pearce (2008).

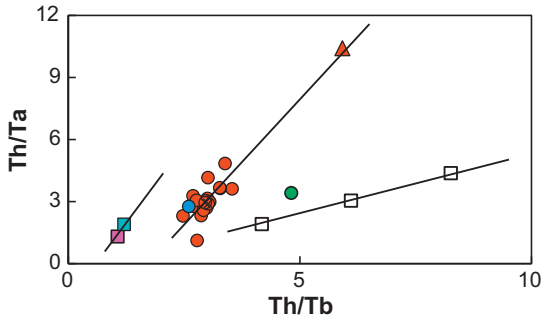


Fig. 9. Bivariate ratio plot of Th/Ta vs. Th/Tb for AKLD, dyke ii, dyke iii and other U–Pb dated dyke samples (symbols as in Fig. 3a).

ratios are also similar, with values of 0.50973 ± 5 and 0.50965 ± 6 for HD-14 and HD-19 respectively, which correspond to $\varepsilon_{Nd}T$ values of −1.7 and −2.9 when compared to CHUR at their respective isochron ages. The negative $\varepsilon_{Nd}T$ values together with the low Nb/Th ratios (compared to primitive mantle values) suggest that this N-S dyke swarm magma originated from a LILE enriched mantle reservoir.

4.3. Paleomagnetism and rock magnetism

At least three specimens per sample were subjected to detailed stepwise AF and/or thermal demagnetization, to a maximum of 150 mT or 600 °C, in order to separate components based on their unblocking temperature. Paleomagnetic results of all fifteen sites are presented in Table 3 and also plotted in Figs. 11 and 12. The directional data were analyzed using Principal Component Analysis (Kirschvink, 1980) in combination with Zijderveld diagrams (Zijderveld, 1967), following standard paleomagnetic procedures. Acceptance criteria for linear segments on vector diagrams were the number of consecutive demagnetization points (≥5) and maximum angle of deviation ≤10° (Kirschvink, 1980). After removing a secondary component with coercivity less than 40 mT, clear

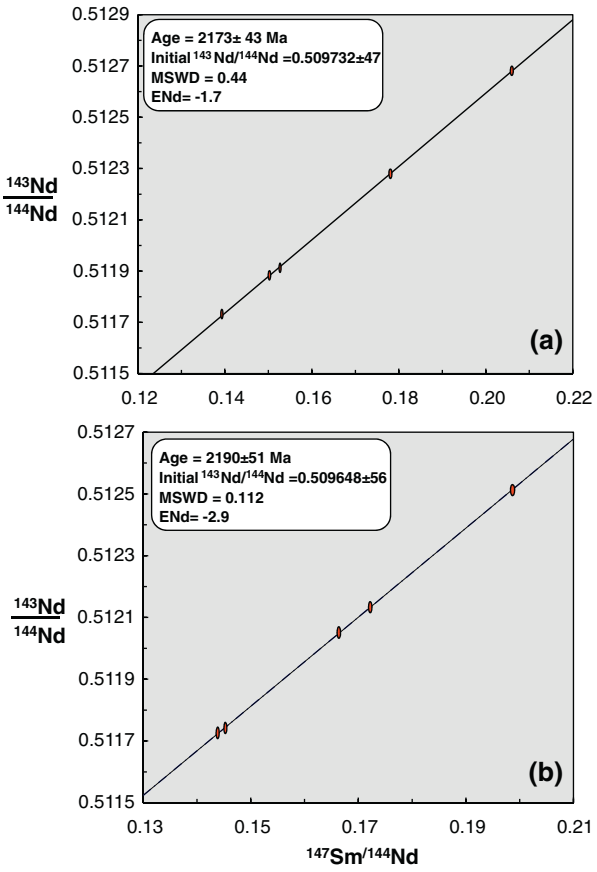


Fig. 10. Sm–Nd whole rock-mineral isochron diagrams for two AKLD samples (a) HD-14 and (b) HD-19 based on data given in Table 2.

linear segments to the origin defined a characteristic component (Fig. 11a–e). Secondary components are generally scattered in most sites, but a poorly defined low coercivity component was recovered from sites C, F, G and H (Fig. 12b), which is similar to the overprinted secondary magnetization recovered from the north eastern parts (around Hyderabad region, Fig. 1) of the older (~2367 Ma) Dharwar giant dyke swarm (Kumar et al., 2012). The origin of this component is unclear. We have also performed on some samples, mixed AF and Thermal demagnetization (Fig. 11f and g). Here again, after having removed a soft component, a segment going through the origin with the same direction as that observed during AF treatment was recovered between 400 °C and 580 °C. The higher coercivity (or higher blocking temperature) component defines a well-grouped characteristic magnetization direction, with west to south-westerly declination and moderately steep up inclination. Within errors, all the sites regardless of their location in the dyke swarm have similar characteristic remanent magnetization directions (Table 3 and Fig. 12). However, the AKLD strike which appears to curve in sympathy with the Archean structural grain of the basement rocks shows a significant ($p = 0.05$) difference in declination between the northern most site I with respect to sites A and B the southernmost sites on it and on dyke ii, and indicate a ~25° counter clockwise rotation in the northern part of AKLD. As against this, neither the declination of the four sites on dyke iii (located north of site I) nor the trend of this dyke is in keeping with this behavior. This inference could have an important implication on the emplacement of the older (2367 Ma) radiating Dharwar giant dyke swarm which was emplaced orthogonal to the regional structural grain in the Archean basement (see Kumar et al., 2012).

Variation of susceptibility with increasing temperature in these samples is illustrated in Fig. 13a and b. Susceptibility values in

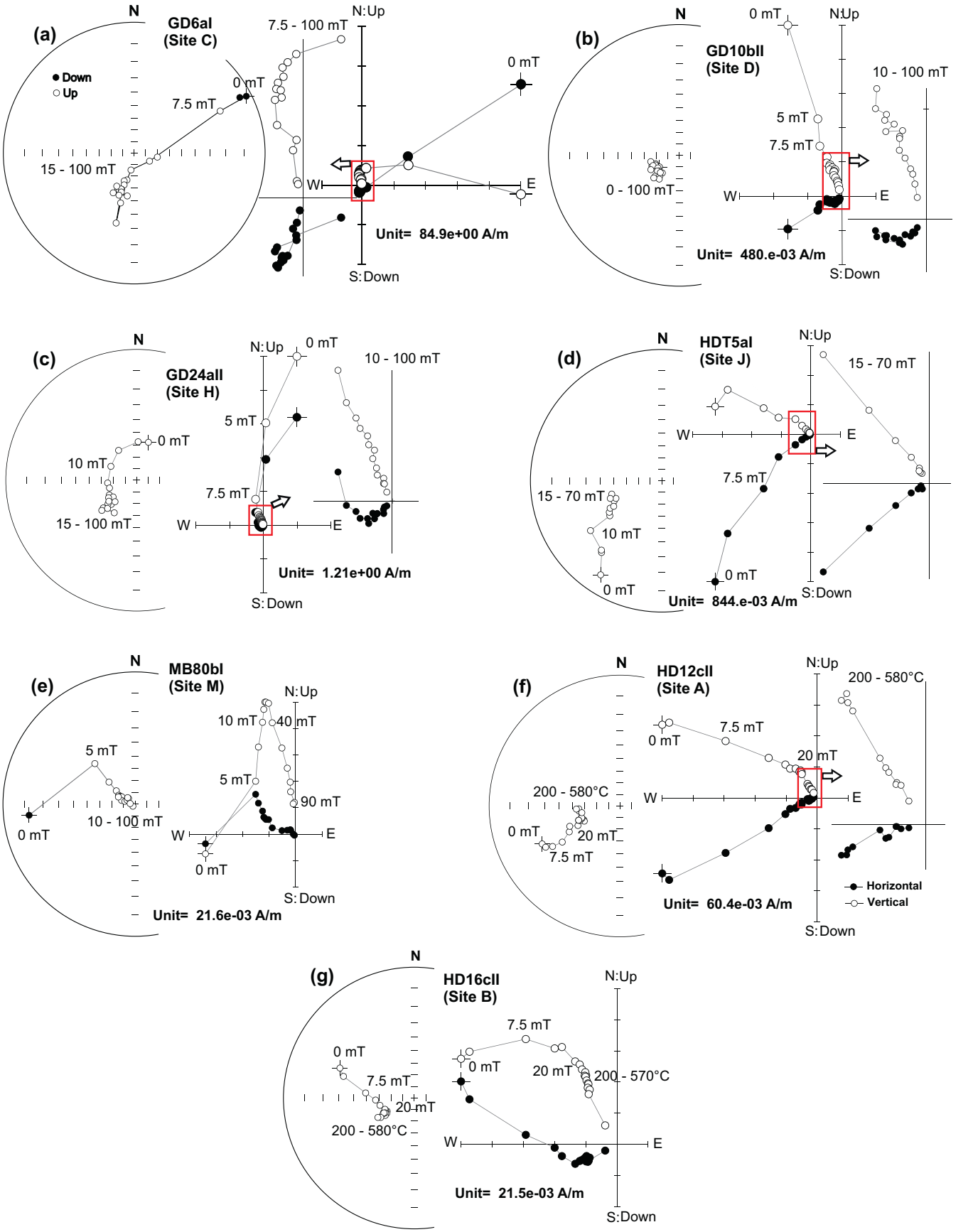


Fig. 11. Zijderveld diagram and equal area stereonet pairs showing the behavior of natural remanence to AF (diagrams a–e) and thermal demagnetization (diagrams f and g) for samples representing several sites. Thermal measurements are in °C and AF measurements are in millitesla (mT). Open/closed circles in the stereoplots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0 plotting and analysis program (Chadima and Hrouda, 2006).

Table 3
Results of paleomagnetic measurements on the Dharwar 2.21 Ga N-S dyke swarm.

Site	Lat. (°N)	Long. (°E)	T	S°	*N*	D_m	I_m	*K*	A_{95}	*p* lat. (°S)	*p* long. (°E)	dp	dm
Component '*P*'													
AKLD													
A	12.767	77.056	43	14	7	243	−55	39	10	30	314	10	14
B	12.851	77.069	125	357	6	245	−63	100	7	27	304	8	11
C	14.216	76.977	96	1	10	204	−60	190	4	56	291	4	5
D	14.218	76.977	143	359	12	218	−66	60	6	43	291	8	9
E	14.209	76.977	172	357	5	220	−55	58	10	48	310	10	14
F	14.204	76.977	157	358	7	240	−59	35	10	32	309	11	15
G	14.117	76.985	107	350	15	231	−55	83	4	40	312	4	6
H	14.112	76.993	104	343	9	236	−63	80	6	34	303	7	9
I	14.778	76.786	89	335	7	215	−65	27	12	47	291	16	19
Mean	13.941	76.977	–	–	9	228	−61	95	5	40	304	6	8
Dyke-ii													
J	12.974	77.353	42	354	6	238	−52	123	6	34	317	6	9
K	12.950	77.378	57	350	5	253	−59	504	3	21	309	4	5
Mean	12.962	77.376	–	–	2	245	−56	–	–	28	313	–	–
Dyke-iii													
L	16.361	77.721	27	357	10	261	−70	104	5	19	295	7	8
M	16.287	77.741	72	4	9	272	−77	111	5	14	284	9	9
N	16.266	77.720	27	5	7	276	−61	101	6	6	305	7	9
O	16.512	77.717	57	13	8	283	−72	47	8	10	281	14	15
Mean	16.357	77.725	–	–	4	273	−72	98	9	12	292	14	16
Mean (all sites)	**14.455**	**77.230**	**–**	**–**	**15**	**239**	**−64**	**43**	**6**	**32**	**302**	**8**	**10**
Component '*S*'													
C	14.216	76.977	96	1	10	70	−5	20	11	–	–	–	–
F	14.204	76.977	157	358	10	27	17	2	44	–	–	–	–
G	14.117	76.985	107	350	13	80	−26	3	30	–	–	–	–
H	14.112	76.993	104	343	10	46	−40	5	24	–	–	–	–
Mean	14.162	76.983	–	–	4	56	−15	6	42	–	–	–	–

Note: Lat. and long are the latitude and longitude in degrees; *T* is the thickness in meters; S° is strike of the dyke at the respective sites. *N* is number of samples studied from each site; D_m is the mean declination; I_m is the mean inclination; *K* is the precision parameter; A_{95} is the radius (°) of the 95% circle of confidence about the mean magnetization direction; *p* lat. and *p* long. are the paleocoordinates of the pole; all statistical parameters are based on sample means. Component '*P*' is inferred as primary component and component '*S*' as of secondary origin.

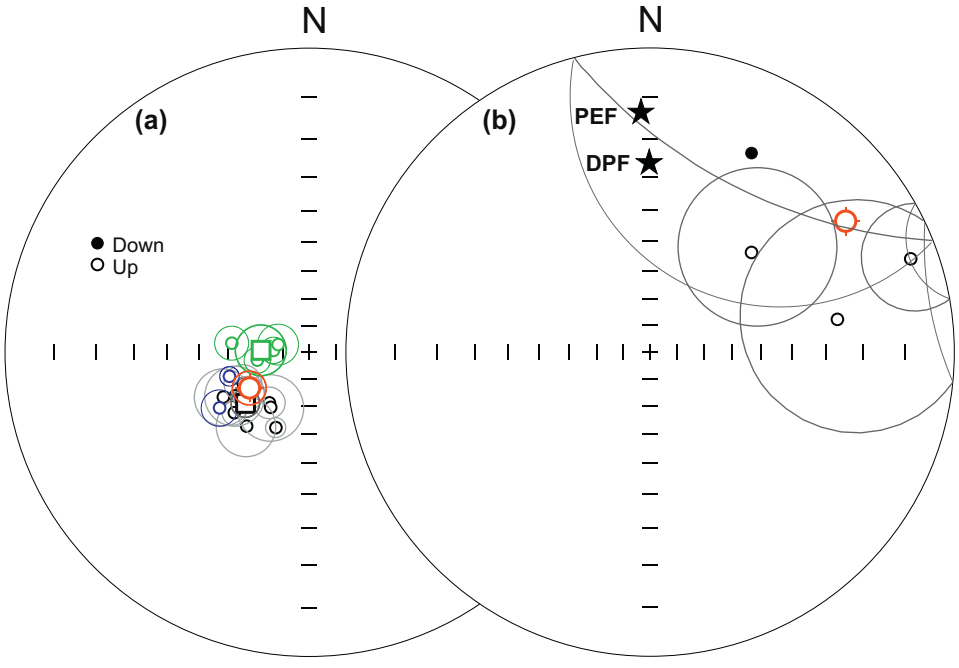


Fig. 12. Stereoplots showing paleomagnetic data. (a) Primary site mean directions with ovals of 95% confidence of all the 15 AKLD, dyke ii and dyke iii sites in black, blue and green respectively. AKLD + dyke ii mean (these two dykes were treated together due to their close proximity) and dyke iii mean with their respective error envelopes are shown as open black and green squares respectively, (b) secondary components are generally scattered in most sites, but a poorly defined component with low coercivity was recovered from sites C, F, G and H. Black stars represent DPF, Dipole field and PEF, Present Earth's field direction based on the 1995 IGRF. Red circles represent grand means in both figures (a) and (b).

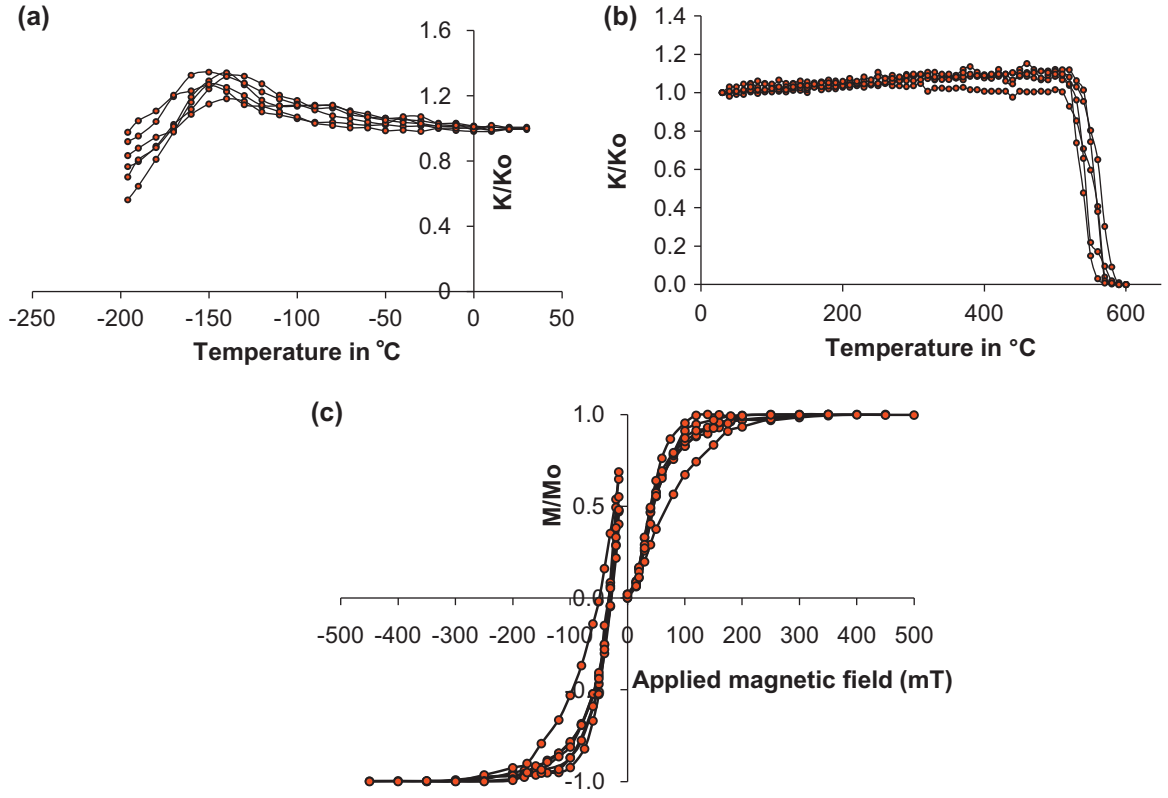


Fig. 13. (a) Variation of magnetic susceptibility from liquid nitrogen temperature (−196 °C) to room temperature and (b) room temperature to 600 °C for representative samples from different sites. (c) Isothermal remanence acquisition curves (acquired magnetization normalized to saturated values vs. applied field in mT).

most of the samples increase gradually from room temperature to about 540–550 °C, by about 5–10%, beyond this temperature the values are seen to decrease to zero close to 580 °C. Low temperature susceptibility variation was determined by cooling samples in liquid nitrogen to −196 °C and recording their susceptibility as the samples warm to room temperature. All samples show a positive anomaly at about −150 °C close to the Verwey Transition for magnetite.

Conventional IRM acquisition studies were also conducted using a pulse magnetiser. Typical saturation curves for these samples are shown in Fig. 13c. The magnetic minerals in them saturate between 150 and 200 mT with coercivity between 40 and 50 mT.

These rocks magnetic studies conducted on whole rock samples clearly suggest that the dominant magnetic mineral is multidomain magnetite. However, careful magnetic experiments on interstitial magnetite grains and on pure plagioclase and pyroxene mineral separates together with electron microscopy studies show the occurrence of elongated <1 μm magnetite inclusions along cleavage planes within the silicate phases pyroxene and plagioclase feldspar (NGRI, IPGP unpublished data). We are inclined to believe that these single domain magnetite grains could be the source of primary remanence in these rocks.

We were unable to obtain baked contact samples during this study to prove the primary nature of the remanence direction obtained here, but several observations suggest that rocks in the study region have not been heated above their blocking temperature (~450–550 °C), after the emplacement of these dykes. Observations supporting this view include: (i) very high coercivities and unblocking temperatures of the grains preserving the characteristic remanence, (ii) consistency of paleomagnetic data between sites despite being separated by up to 450 km, (iii) absence of metamorphic events after ~2500 Ma as evidenced by the consistency of titanite and zircon ages in the country rocks (both yielding ~2500 Ma ages) around Hutti, Kolar and Ramagiri (summarised by Anand and Balakrishnan, 2010) despite the marked difference in the blocking temperatures of titanite (600–650 °C) and zircon (~900 °C) Cherniak (1993) and also by several ~2450 Ma biotite-whole rock Rb-Sr two point isochron ages (biotite has a much lower closure temperature of ~350 °C) for granites and gneisses in the region. One of these biotite age determinations (2489 ± 35 Ma) is less than 15 km from sites A, B, J and K (Krishna, 2007), and (iv) preservation of primary remanence in the older 2367 Ma dominant dyke swarm in this region, as inferred from a positive baked contact test (Halls et al., 2007; Kumar and Bhalla, 1983). Thus we are of the opinion that the nature of magnetization recorded by AKLD and the other N-S dykes reported here is primary acquired at the time of dyke emplacement.

Except for the 2367 Ma dyke swarm, the steep negative inclination magnetization direction of these N-S dykes was not found in any other Paleoproterozoic dyke swarms in the Dharwar craton. However, a similar paleomagnetic direction as this has been reported from charnokites (Neoarchean) near Vandallore (Mondal et al., 2009) within the southern granulite province located about 30 km southwest of Chennai and more than 300 km south east of our sampling sites. This direction was interpreted as acquired at the time of uplift cooling of the charnokite rocks.

To use the paleomagnetic data obtained from the N-S dyke swarm in this study for paleocontinental reconstructions, it is mandatory to ascertain if the Virtual Geomagnetic Pole (VGP) has averaged out Secular Variation (SV). The calculated VGP scatter value (Cox, 1970) for the fifteen sites studied on the N-S dykes is 16.9° + 2.3°/−5.4° and is similar to the observed VGP angular dispersion value of about 16° (Johnson et al., 2008; McElhinny and McFadden, 1997) at the same latitude (Dharwar N-S dyke swarm paleolatitude of 46°) during the last 0–5 Ma period, suggesting that SV was adequately averaged out.

Table 4
Palaeomagnetic data used for reconstruction in Fig. 14.

Age (Ma)	S_{lat} (°N)	S_{long} (°E)	P_{lat} (°N)	P_{long} (°E)	dp	dm	Formation	Authors
2231 ± 2	64.5	−110.3	−51	310	6	8	Malley dykes (Slave)	Buchan et al. (2012)
~2190	62.5	−99	−1	122	6	10	Tulemalu (Rae)	Fahrig et al. (1984)
2210 ± 10	14.5	77	−32	302	8	10	AKLD, dykes ii and iii (Dharwar)	This paper
2216 + 8/−4	48.4	−79	15	104	4	7	Senneterre (Superior)	Buchan et al. (1993)

S_{lat} and S_{long} are site latitude (°N) and site longitude (°E) respectively. P_{lat} and P_{long} are pole latitude (°N) and pole longitude (°E) of the Virtual Geomagnetic Poles; dp, dm are the semi-axes of the oval of 95% confidence ellipse in degrees for the VGP. The primary nature of the remanence direction obtained on the Tulemalu dykes is yet to be established.

5. Discussion

5.1. Age of Andhra-Karnataka long dyke

We interpret that the Sm–Nd isochron ages of 2173 ± 43 Ma and 2190 ± 51 Ma obtained on the two samples from this dyke are close to its emplacement age, within errors. Within errors these ages are similar to four recently reported Paleoproterozoic dykes in the Dharwar craton (French and Heaman, 2010). These include a N-S striking dyke at Kandlamadugu town with an age of 2220.5 ± 4.9 Ma and a 2209.3 ± 2.8 Ma NNW-SSE trending dyke near Somala town, both located at the south eastern corner of the eastern Dharwar craton, about 150 km east of the AKLD (Fig. 1). About 400 km north east of the AKLD and ~200 km from dyke iii in the same direction, an E-W trending dyke near Bandepalem yielded a 2176.5 ± 3.7 Ma age, and a NW-SE striking dyke located about 160 km North West of AKLD near Dandeli has an age of 2180.8 ± 0.9 Ma (see Fig. 1). The latter two dykes were inferred to be part of a large radiating dyke swarm emplaced 2181–2176 Ma ago (French and Heaman, 2010). Geochemical comparison of all these dykes presented above suggests that Kandlamadugu is chemically coherent with the ALKD, dyke ii and dyke iii. Similarity in their trends (NNW) also suggests these dykes could belong to the same swarm. This inference is consistent with a preliminary U–Pb baddeleyite age estimate of 2215 Ma, for the AKLD (personal communication by M. A. Hamilton in Bleeker and Jayananda, 2010) which is similar to Kandlamadugu U–Pb baddeleyite age of 2220.5 ± 4.9 Ma, within error. This N-S to NNW-SSE trending dyke swarm (Fig. 1) could therefore encompass a much larger aerial extent than shown earlier (French and Heaman, 2010). Its present dimension suggests it is one of the predominant dyke swarms in the Eastern Dharwar craton.

5.2. Global incidence of the ~2210 Ma event

Mafic magmatism at 2210 ± 20 Ma (only U–Pb ages are considered here) has a wide distribution and occurs in several continental regions including North America, Greenland, Europe, Africa, Australia and India. The impact of this event was most prolific in North America which includes Klotz (2209 ± 1 Ma; Buchan et al., 1998), Senneterre dykes (2216 + 8/−4 Ma; Buchan et al., 1998), Nipissing sills (2217 ± 4 Ma; Noble and Lightfoot, 1992), and a few smaller dyke swarms including Anuc (2220 ± 1 Ma), Kogaluk bay (2212 ± 3 Ma) and Couture (2199 ± 5 Ma) all within the Superior province (Maurice et al., 2009). The Malley (2231 ± 2 Ma; Buchan et al., 2012), MacKay dykes (2208 ± 5 Ma; LeCheminant et al., 1996), Blachford Lake Intrusion (2186 ± 10 Ma; Wanless et al., 1979) and Dogrib dykes (2188 ± 4 Ma; LeCheminant et al., 1997) in the Slave province and the Tulemalu–MacQuoid dyke swarm (2190 Ma; LeCheminant et al., 1997) in the Rae province are all roughly the same age. Coeval events reported from the Greenland shield are the BN-1 (Isukasia) dykes (2214 ± 10 Ma; Nutman et al., 1995), in the Fennoscandian shield, the Koli event which includes the Pera Pohja (2200 Ma; Perttunen, 1991), Lapland (2220 ± 11 Ma; Tyrvainen, 1983) and Kuusamo intrusives (2206 ± 9 Ma; Silvennoinen, 1991) and coeval volcanism at Cheela Springs (2209 ± 15 Ma; Martin et al., 1998) from the Hamersly province, NW Australia. Kandlamadugu, Somala (French and Heaman, 2010) and AKLD swarm from the Dharwar craton when added to this long list, indicate wide spread contemporaneous mafic magmatism at ~2210 Ma that may be linked to enhanced mantle plume activity, resulting in the breakup of Archean supercontinents during this time (LeCheminant et al., 1997; Buchan et al., 1998).

Table 5
Euler poles used for reconstruction.

Age (Ma)	Fit	λ (°N)	Φ (°E)	ω (°)
Dharwar to superior				
2210 ± 10	Fit "S"	43.4	355.8	151.6
2210 ± 10	Fit "N"	−26.9	193.5	27.6
Slave to superior				
2231 ± 2	Fit "S"	−27.4	180.2	45.3
Rae to superior				
~2190	Fit"S"	−61.1	32.2	26.9

λ and Φ are the latitude and longitude of Euler poles respectively. ω is the Euler angle used for rotation. Euler angle (ω) is positive for counter-clockwise rotation and negative for clockwise rotation.

5.3. Paleoreconstruction

Though the ~2210 Ma event has a worldwide distribution as summarized above, reliable paleomagnetic data together with precise age determinations are limited to only a few cratons. These include the Senneterre dykes of the Superior, Malley dykes from the Slave and the Tulemalu dykes from Rae cratons, besides the N-S dyke swarm from the Dharwar. Our paleoreconstructions are therefore limited to only these cratons (Table 4).

When plotted (Fig. 14, using PaleoMac plotting program of Cogne, 2003) in their ancient latitudes and azimuthal orientations as originally reported (Table 4), Dharwar at ~2210 Ma is located at intermediate latitudes (46°) in the southern hemisphere, Superior, Slave and Rae cratons also plot in the same hemisphere but at shallower latitudes of 27°, 34° and 20° respectively.

There could be several possibilities of assembling these cratons, however, our approach was to plot Dharwar, Slave and Rae with respect to the Superior craton, as it has a fairly well defined Paleoproterozoic Apparent Polar Wander Path (APWP, Buchan et al., 2007). Dharwar, Slave and Rae were rotated about their respective Euler poles (Euler pole coordinates and the degree of rotation are given in Table 5) resulting in a reconstruction that is illustrated in Fig. 14. In this configuration (enlarged inset in Fig. 14), which is in sharp contrast with that proposed recently by French and Heaman (2010), the Somala and Dandeli dykes from the Dharwar, Malley, Dogrib and Mackay dykes from the Slave, Tulemalu-MacQuoid from the Rae and Klotz dykes from the Superior craton though strictly not coeval are all approximately (within 30°) parallel to each other. The AKLD, dyke ii, dyke iii and Kandlamadugu appear to be parallel to the Anuc and perpendicular to the Senneterre and Biscotasing dyke swarms in the Superior craton. Though all these swarms were emplaced within a short time span, a critical evaluation of not only

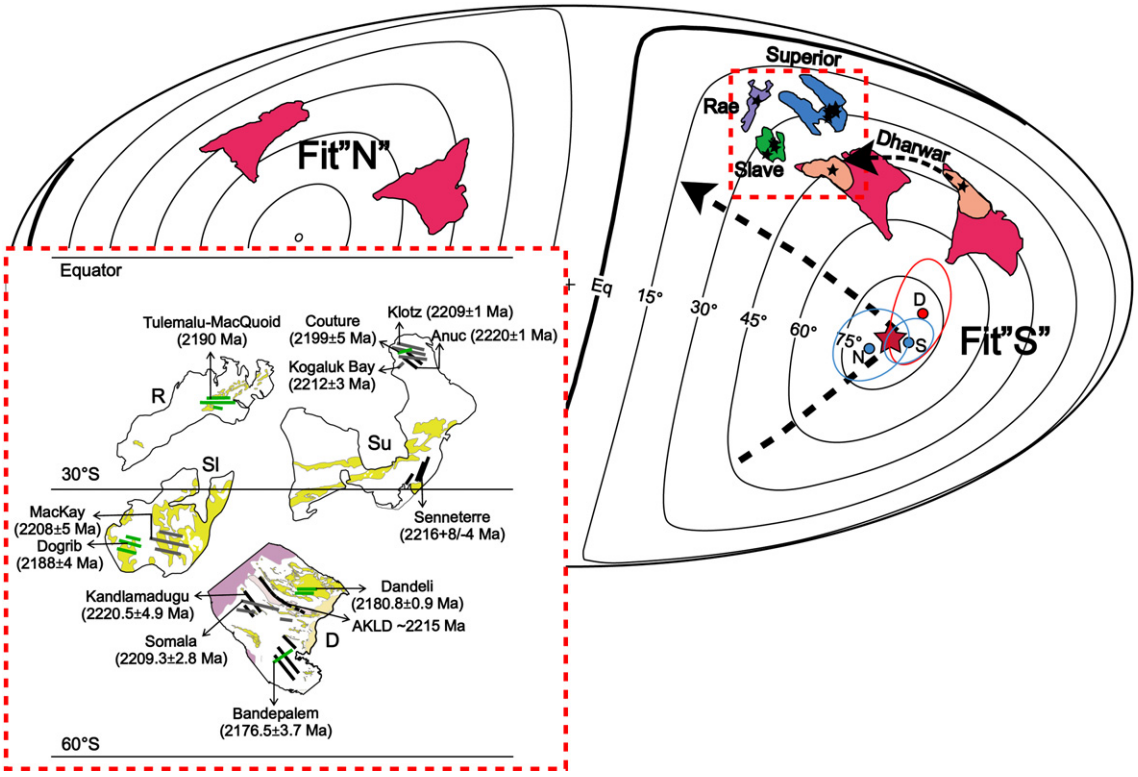


Fig. 14. Mollweide projection showing paleopositions of the North American cratons Superior (in blue), Slave (green), Rae (purple), and the Dharwar craton (in brown) at ~ 2210 Ma. This reconstruction was done with respect to Superior craton. Assuming that the field was a geocentric axial dipole, Earth's rotation axis for this reconstruction is based on the proximity of the 2210 Ma poles from Superior craton and is shown as a red star. Blue dots represent Senneterre (S), Nipissing (N; Stupavsky and Symons, 1981) and Dharwar (D) VGPs (Table 4). Dashed line represents the Apparent Polar Wandering Path (APWP) for the Superior craton (after Buchan et al., 2007). Thick black line in the figure represents the paleo-equator (Eq.) and thin lines are the latitude grids, spaced at 15°. Black stars: Paleomagnetic sampling sites. Figure also illustrates several alternate positions for India when rotated along its paleolatitude, both in the southern and northern hemispheres due hemispheric ambiguity. This reconstruction was done using "PaleoMac" computer program (Cogne, 2003). Inset is an enlarged view of the reconstructed cratons at ~2210 Ma to display dyke swarm orientation (shown in Galls projection using Torsvik and Smethurst, 1999, plotting program). Coeval ~2210 Ma dyke swarms are represented as gray and black thick lines and the marginally younger (~2190 Ma) swarms are shown as green lines. Outlines of Neoarchean sequences are also shown within the cratons in light green for comparison of their regional structural grain.

the swarm geometry but also their geochemical characteristics is essential for establishing inter swarm correlation across cratons to infer their tectonic setting.

Regarding the ancestry of the four cratons in this assembly, based on the degree of geological similarity, Superior was inferred to be part of the Superia supercraton, but that of the Rae craton was unknown (Bleeker, 2003; Mueller et al., 2005). Dharwar and Slave cratons were believed to be the progeny of the Supercraton Sclavia (that existed between 2.6 and 2.2 Ga; Bleeker, 2003) as they were characterized by strikingly similar Archean histories with several corelatable features, including coeval large scale granite emplacement (granite bloom) at ca. 2.60–2.58 Ga, which marked the end of cratonization in these granite–greenstone terrains. In our reconstruction at ~2.21 Ga, though Dharwar, Slave, Superior and Rae are shown to be juxtaposed, with common mafic dyke events at ~2.21 and 2.18 Ga (French and Heaman, 2010), the conspicuous absence of the 2.367 Ga Dharwar giant dyke swarm (Kumar et al., 2012) in the other three cratons and with their Neoarchean structural grain being nearly perpendicular to each other, suggests this assembly may not be valid during the Paleoproterozoic-Neoarchean times. In which case, neither Superia nor Sclavia could have been an ancestral supercraton for the Dharwar. Paleomagnetic data from the 2.367 Ga old Dharwar giant dyke swarm (Kumar et al., 2012) shows that Dharwar was situated at polar latitudes (~69°S) at that time. The only other craton known to be located at steep latitudes (~50°S) during the Paleoproterozoic time is the Yilgarn, as inferred from the ~2400 Ma Widgiemooltha dyke swarm paleomagnetic data (Evans, 1968; Smirnov and Evans, 2006; Nemchin and Pidgeon, 1998; Doehler and Heaman, 1998). Based on this evidence a common supercraton ancestry for both Dharwar and Yilgarn cratons was proposed (Halls et al., 2007). The inferred Archean supercraton 'Ur' could be a likely candidate (Rogers and Santosh, 2003).

6. Conclusions

The remarkable chemical homogeneity observed all along the dyke length of AKLD could be due to limited fractionation, rapid emplacement and magma derivation from a single homogeneous source rather than from multiple sources. Both elemental and isotopic compositions also suggest that this source was enriched relative to Bulk Earth or the CHUR.

Occurrence of the ~2210 Ma event in the Dharwar and in at least six other cratons suggests it has a truly global distribution, which could indicate large scale mantle perturbation and re-organization of the mantle convection regime during this time, which could have resulted in the breakup of Archean supercratons (LeCheminant et al., 1997; Buchan et al., 1998).

Of the several possible Paleoreconstructions at ~2210 Ma, one shows that the four cratons Dharwar, Slave, Superior and Rae could have been juxtaposed at intermediate to shallow latitudes (20–46°) at this time. However, lack of reliable Neoarchean paleomagnetic data from the Dharwar inhibits tracing its ancestry to either of the Neoarchean Supercratons: Sclavia or Superia.

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